SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)*

Assessment Tool Weightage: 50%

Dr. G. A. Senthil

QUESTIONS: 10 **Total Marks: 50**

Annexure A

RL Basic Experiments demo with Keras and TensorFlow using Anaconda navigator

Duration: 2hrs

Experiment 1: Installation of Reinforcement Learning Development Environment

Aim: To install and configure Anaconda Navigator, Python, TensorFlow, Keras, Gymnasium (OpenAI Gym), NumPy, Matplotlib, and other required libraries, and verify the environment by executing a simple Reinforcement Learning program.

Experiment 2: Familiarization with OpenAI Gym/Gymnasium Environments

Aim: To create, initialize, and interact with standard Reinforcement Learning environments such as FrozenLake, CartPole, and MountainCar using Gymnasium to understand the concepts of states, actions, rewards, and episodes.

Experiment 3: Implementation of the Reinforcement Learning Framework

Aim: To implement the basic Reinforcement Learning framework by designing an agent that interacts with an environment through states, actions, rewards, and policies using Python.

Experiment 4: Simulation of a Markov Decision Process (MDP)

Aim: To develop a simple Markov Decision Process model and simulate state transitions, reward functions, and policy execution for understanding sequential decision-making problems.

Experiment 5: Bellman Equation Demonstration

Aim: To implement Bellman Expectation and Bellman Optimality Equations for calculating state-value and action-value functions and analyze the convergence of value estimation.

Experiment 6: Exploration vs. Exploitation Strategies

Aim: To implement ε-Greedy, Greedy, and Random action-selection strategies and compare their performance in balancing exploration and exploitation during Reinforcement Learning.

Experiment 7: Multi-Armed Bandit Problem

Aim: To implement the Multi-Armed Bandit problem using ε-Greedy and Upper Confidence Bound (UCB) algorithms and evaluate cumulative rewards obtained by different action-selection methods.

Experiment 8: Reward Visualization in Reinforcement Learning

Aim: To simulate an RL agent in a simple environment and visualize cumulative rewards, episode rewards, and learning performance using Matplotlib graphs.

Experiment 9: TensorFlow and Keras-Based Neural Network for RL

Aim: To build a simple feed-forward neural network using TensorFlow and Keras for approximating value functions in Reinforcement Learning environments.

Experiment 10: Mini Reinforcement Learning Project Using CartPole

Aim: To develop a basic Reinforcement Learning agent using TensorFlow and Keras to solve the CartPole environment and evaluate its learning performance through episode rewards and success rate.

EXPERIMENT-1

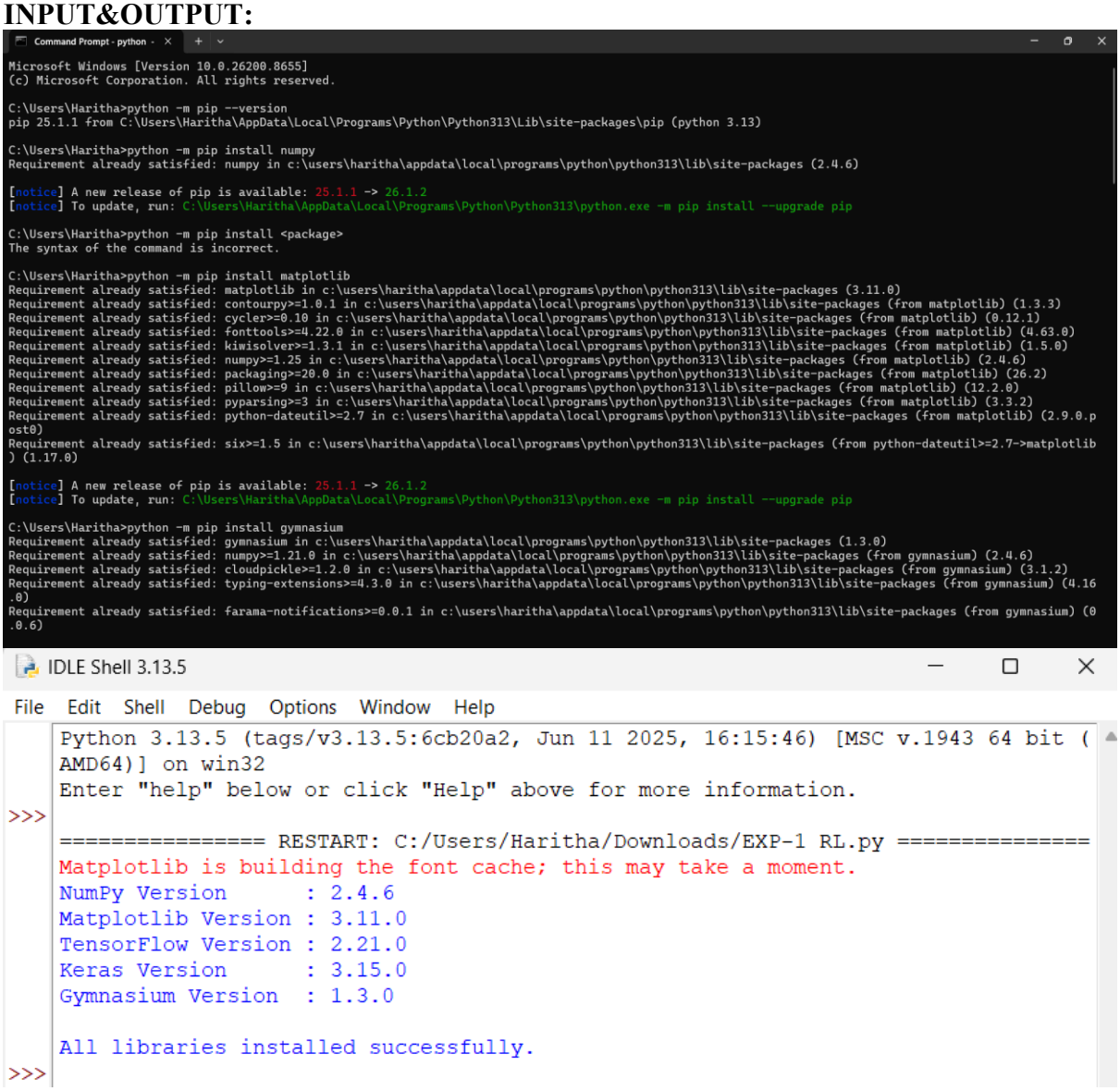
AIM:

To install and configure Anaconda Navigator, Python, TensorFlow, Keras, Gymnasium (OpenAI Gym), NumPy, Matplotlib, and other required libraries, and verify the environment by executing a simple Reinforcement Learning program.

PROGRAM:

```
import numpy
import matplotlib
import tensorflow
import keras
import gymnasium
print("NumPy Version      :", numpy.__version__)
print("Matplotlib Version :", matplotlib.__version__)
print("TensorFlow Version :", tensorflow.__version__)
print("Keras Version      :", keras.__version__)
print("Gymnasium Version  :", gymnasium.__version__)
print("\nAll libraries installed successfully.")
```

INPUT&OUTPUT:



EXPERIMENT -2

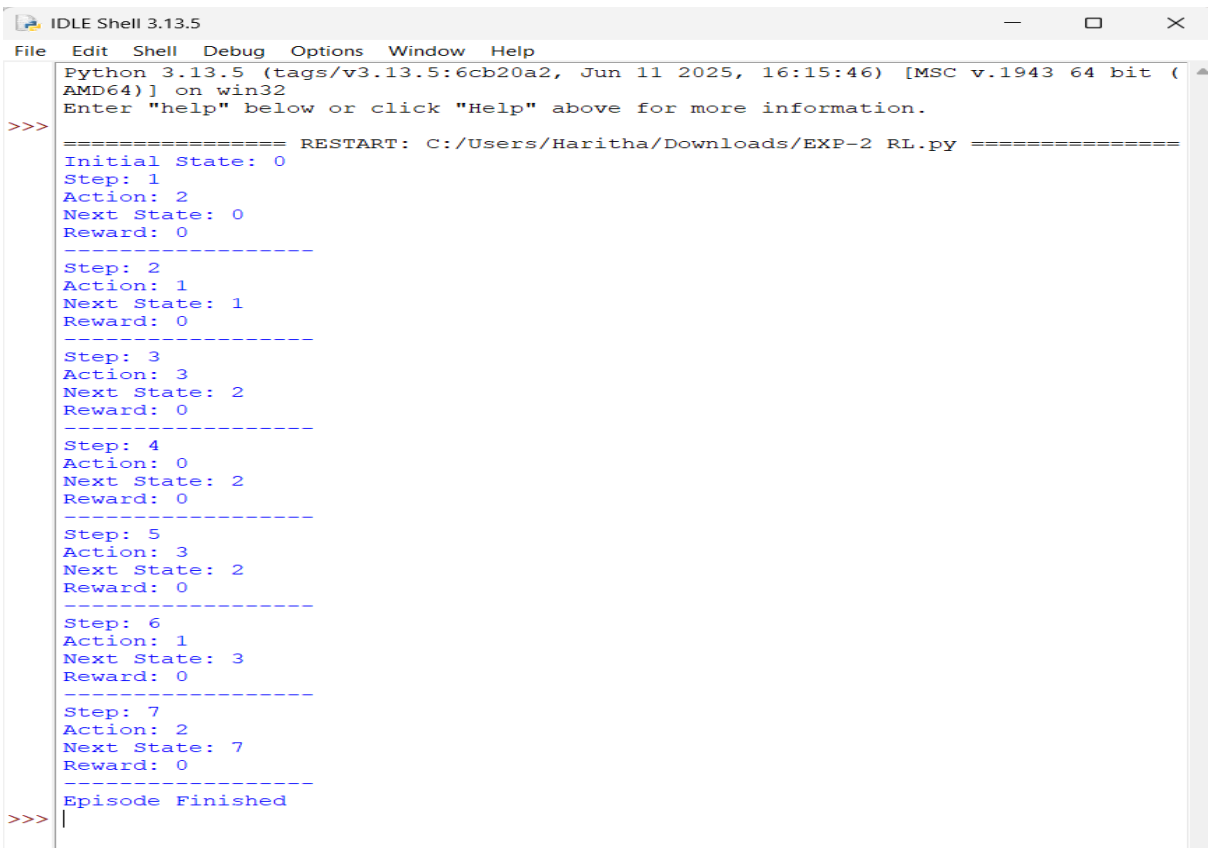
AIM:

To create, initialize, and interact with standard Reinforcement Learning environments such as FrozenLake, CartPole, and MountainCar using Gymnasium to understand the concepts of states, actions, rewards, and episodes.

PROGRAM:

```
import gymnasium as gym
# Create the FrozenLake environment
env = gym.make("FrozenLake-v1")
state, info = env.reset()
print("Initial State:", state)
for step in range(10):
    print("Next State:", next_state)
    print("Reward:", reward)
    print("------------------") state = next_state
    if terminated or truncated:
        print("Episode Finished")
        break
env.close()
```

INPUT&OUTPUT:



EXPERIMENT-3

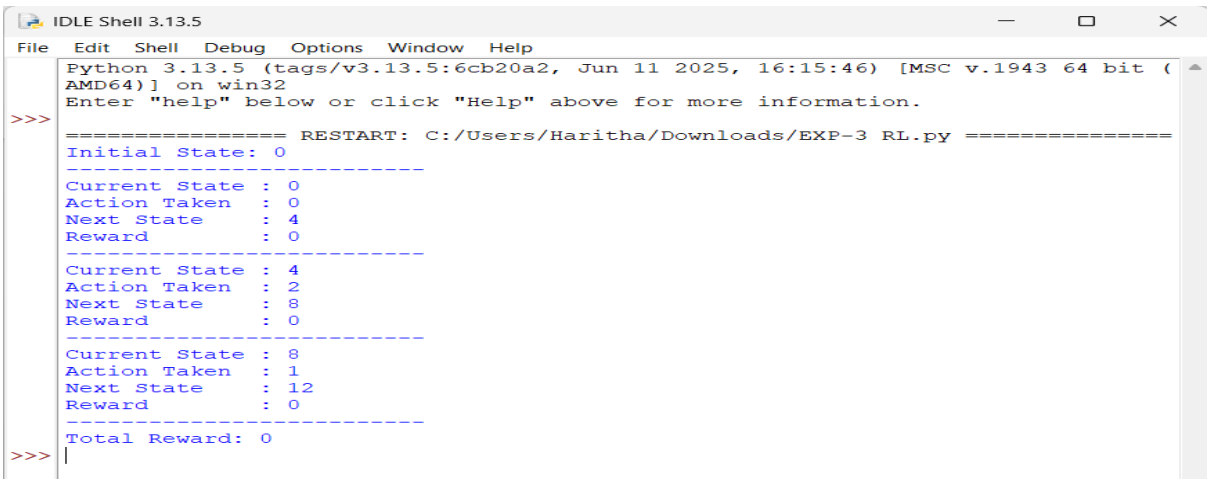
AIM:

To implement the basic Reinforcement Learning framework by designing an agent that interacts with an environment through states, actions, rewards, and policies using Python.

PROGRAM:

```
import gymnasium as gym
env = gym.make("FrozenLake-v1")
state, info = env.reset()
print("Initial State:", state)
total_reward = 0
done = False
while not done:
    action = env.action_space.sample()
    next_state, reward, terminated, truncated, info = env.step(action)
    print("--------------------------")
    print("Current State :", state)
    print("Action Taken  :", action)
    print("Next State    :", next_state)
    print("Reward        :", reward)
    state = next_state
    total_reward += reward
    done = terminated or truncated
print("--------------------------")
print("Total Reward:", total_reward)
env.close()
```

OUTPUT:



EXPERIMENT-4

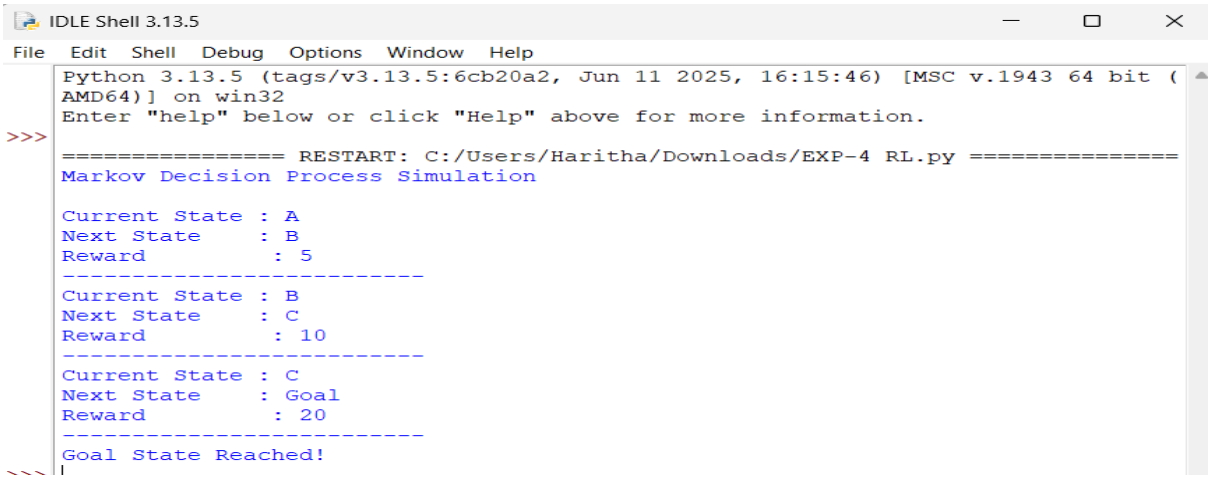
AIM:

To develop a simple Markov Decision Process model and simulate state transitions, reward functions, and policy execution for understanding sequential decision-making problems.

PROGRAM:

```
states = ["A", "B", "C", "Goal"]
# Policy (State Transitions)
policy = {
   "A": "B",
   "B": "C",
   "C": "Goal"
}rewards = {
   ("A", "B"): 5,
   ("B", "C"): 10,
   ("C", "Goal"): 20
}current_state = "A"
print("Markov Decision Process Simulation\n")
while current_state != "Goal":
   next_state = policy[current_state]
   reward = rewards[(current_state, next_state)]
   print("Current State :", current_state)
   print("Next State    :", next_state)
   print("Reward        :", reward)
   print("--------------------------")
   current_state = next_state
print("Goal State Reached!")
```

OUTPUT:



EXPERIMENT-5

AIM:

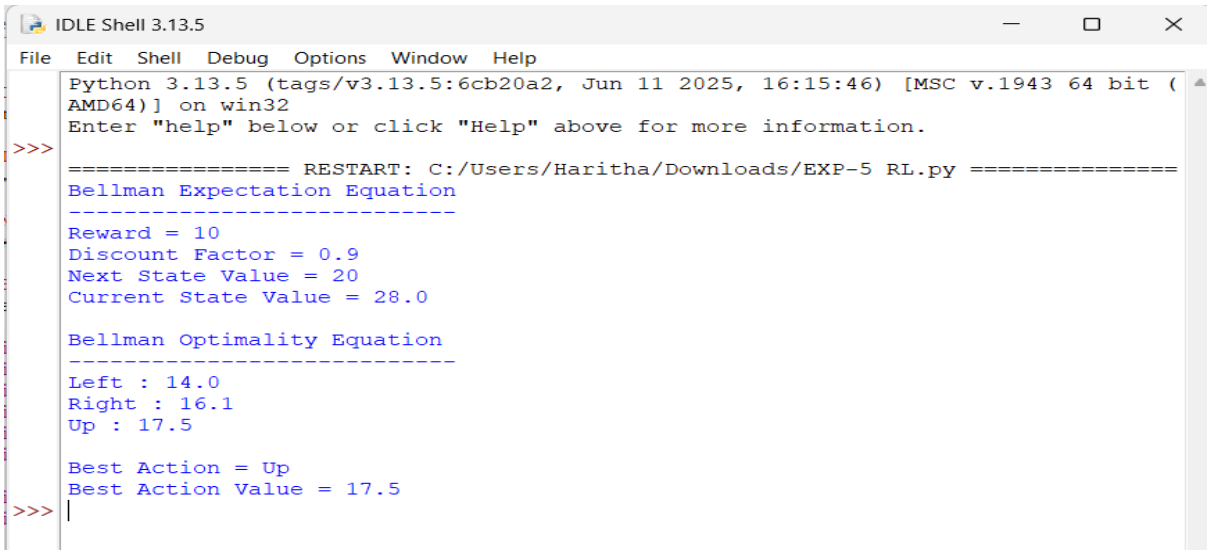
To implement Bellman Expectation and Bellman Optimality Equations for calculating state-value and action-value functions and analyze the convergence of value estimation.

PROGRAM:

```
# Discount Factor
gamma = 0.9
# Immediate Reward
reward = 10
# Next State Value
next_state_value = 20
# Bellman Expectation Equation
state_value = reward + gamma * next_state_value
print("Bellman Expectation Equation")
print("----------")
print("Reward =", reward)
print("Discount Factor =", gamma)
print("Next State Value =", next_state_value)
print("Current State Value =", state_value)
print("\nBellman Optimality Equation")
print("----------")
# Action Values
actions = {
    "Left": 5 + gamma * 10,
    "Right": 8 + gamma * 9,
    "Up": 4 + gamma * 15
```

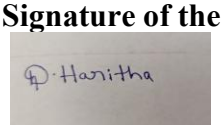
```
}
for action, value in actions.items():
    print(action, ":", value)
best_action = max(actions, key=actions.get)
print("\nBest Action =", best_action)
print("Best Action Value =", actions[best_action])
```

OUTPUT:



Code of Conduct:

I **Dasari Haritha(192425254)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand